Skin Cancer Classification using Deep Learning

Gowthami Ratikrinda
University of Michigan
gratikri@umich.edu

Harshitha Kundarapu
University of Michigan
harshitk@umich.edu

***Abstract*—Skin cancer is a common and potentially fatal disease, where early and accurate diagnosis is critical. Visual inspection of skin lesions by dermatologists is subjective and can lead to inter-observer variability. To address this, automated classification systems are developed to improve consistency and reliability in skin lesion diagnosis. This paper studies multi-class dermoscopic image classification using the HAM10000 dataset (10,015 images, 7 classes). We compare three models: a custom three-block CNN trained from scratch, and two transfer learning models, ResNet-50 and EfficientNet-B3 pre-trained on ImageNet. To handle class imbalance, we use lesion-level data splitting, oversampling to 500 images per minority class, and weighted cross-entropy loss. On a held-out test set of 1,516 images, EfficientNet-B3 achieves the best performance (Macro F1: 0.6709, ROC-AUC: 0.9509), while ResNet-50 achieves the highest accuracy (76.12%). The custom CNN reaches 42.61% accuracy, highlighting the effectiveness of transfer learning for imbalanced medical image datasets.**

***Index Terms*—Index Terms— Skin cancer classification, dermoscopic images, HAM10000 dataset, deep learning, convolutional neural networks (CNN), transfer learning, ResNet-50, EfficientNet, class imbalance, medical image analysis.**

I. Introduction

A. Problem Statement

Skin cancer accounts for a large proportion of global cancer cases. Melanoma, the most severe type, has a high survival rate if detected early. However, visual diagnosis is difficult due to the similarity between benign and malignant lesions, even for experienced dermatologists. Dermoscopy improves accuracy but still depends on expert interpretation. The HAM10000 dataset (10,015 images, 7 classes) reflects a real-world challenge of class imbalance. The nevus (nv) class dominates the dataset (6,705 samples, 66.9%), while classes such as dermatofibroma (df, 115 samples) and vascular lesions (vasc, 142 samples) are under-represented. Building automated systems that can classify all seven categories, including melanoma (mel) and basal cell carcinoma (bcc), can improve consistency, support clinical decisions, and reduce diagnostic cost.

B. Why classification works?

This problem is a supervised multi-class image classification task. Each dermoscopic image has a corresponding ground-truth diagnosis label from one of seven categories, making standard supervised classification the natural methodological choice. Convolutional neural networks are well-suited here because they exploit spatial hierarchies in image data — early layers detect edges and textures, while deeper layers learn high-level semantic features such as asymmetry, border irregularity, and color variation that are diagnostically meaningful for skin lesions.

TABLE I
Distribution of Diagnostic Classes in HAM10000

Class Code	Diagnostic Label	Sample Count (n)	Percentage (%)
`nv`	Melanocytic nevi	6,705	66.95%
`mel`	Melanoma	1,113	11.11%
`bkl`	Benign keratosis-like lesions	1,099	10.97%
`bcc`	Basal cell carcinoma	514	5.13%
`akiec`	Actinic keratoses	327	3.27%
`vasc`	Vascular lesions	142	1.42%
`df`	Dermatofibromas	115	1.15%
Total		**10,015**	**100.00%**

C. Related Work

Transfer learning has emerged as the dominant approach for HAM10000 classification. Bisla et al. [3] demonstrated that EfficientNet-B0 outperforms ResNet-50 on this dataset despite fewer parameters, motivating our choice of EfficientNet-B3. Nasr-Esfahani et al. [4] evaluated five architectures on HAM10000, with DenseNet121 achieving 94.29% and ResNet-50 reaching 87.10%, highlighting the difficulty of minority class recognition without explicit imbalance handling.

Class imbalance has been consistently identified as the core challenge. Ali et al. [5] and Vuong et al. [6] both showed that combining oversampling with weighted loss significantly improves rare class recall. Our work extends these findings through a controlled three-way comparison under identical preprocessing conditions, using lesion-level splitting and Macro F1 as the primary clinical metric.

II. METHODOLOGY

A. Dataset Summary

The HAM10000 dataset was sourced from Kaggle [1]. It consists of 10,015 dermoscopic images categorized into seven diagnostic classes. As summarized in Table I, the dataset exhibits a pronounced class imbalance: melanocytic nevi (nv) account for 66.95% of the total samples, whereas dermatofibromas (df) represent only 1.15%. This skewness necessitates specific preprocessing strategies, such as oversampling or weighted loss functions, to ensure the model does not develop a bias toward the majority class.

B. Data Preprocessing

To ensure model robustness and prevent overfitting to the majority class, a multi-stage preprocessing pipeline was implemented as follows:

1) **Data Leakage Prevention:** To avoid optimistic bias, the dataset was partitioned at the lesion level rather than the image level. Using a 70/15/15 ratio (random seed 42), images associated with the same `lesion_id` were kept within the same split. This resulted in 6,987 samples for training, 1,512 for validation, and 1,516 for testing.
2) **Imbalance Mitigation:** A hybrid strategy was employed to address the heavy skew in class distribution:

   - *Oversampling: Oversampling:* Minority classes in the training set were oversampled with replacement [2] to reach a minimum threshold of 500 images per class, resulting in a balanced training pool of 8,237 images.
   - *Cost-Sensitive Learning:* Inverse class frequency weights were calculated using the formula $w_j = \frac{N}{k \cdot n_j}$. Weights ranged from 0.2111 (nv) to 13.3086 (df) and were integrated into the Cross-Entropy loss function to penalize minority class errors more heavily.
   - The threshold of 500 images per minority class was selected as a balance between two competing risks. Full equalization to the majority class level (4,694 training samples for `nv`) would introduce severe overfitting on minority classes due to excessive repetition of the same images [2]. Conversely, leaving the distribution entirely uncorrected would cause the model to overwhelmingly predict `nv`, as confirmed by our Custom CNN baseline results (weighted precision 0.75 but recall 0.43). The 500-sample floor was chosen to provide sufficient class representation while limiting duplication artifacts. The residual imbalance is further addressed by the weighted cross-entropy loss, which assigns a penalty of up to 13.3× for misclassifying the rarest class (`df`), ensuring the loss function remains sensitive to minority pathologies even when their sample count is lower than `nv`.

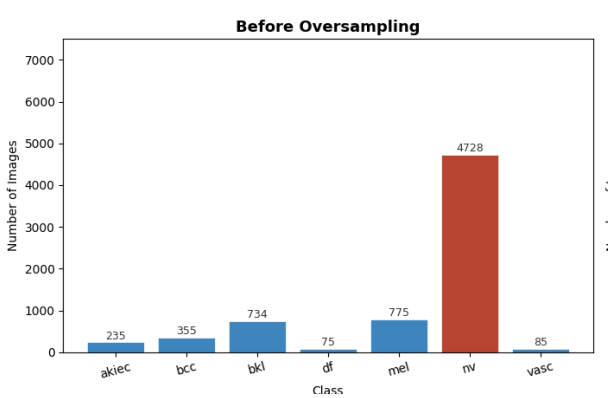

Fig. 1. Training set class distribution before oversampling.

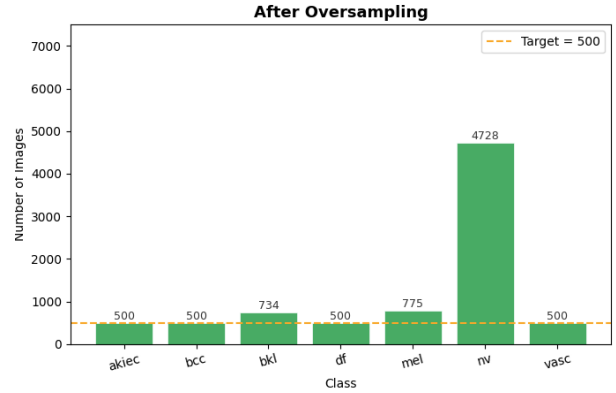

Fig. 2. Training set class distribution after oversampling.

3) **Stochastic Augmentation:** To improve generalization, the training set underwent real-time transformations including horizontal/vertical flips, 30° rotations, and color jittering (brightness/contrast/saturation). These augmentations were excluded from the validation and test sets to prevent distribution shift during evaluation.
4) **Standardization:** All images were resized to 300×300 pixels to match the native input resolution of EfficientNet-B3 [8]. Pixel values were normalized using the ImageNet [9] mean ([0.485, 0.456, 0.406]) and standard deviation ([0.229, 0.224, 0.225]) to facilitate transfer learning with pre-trained architectures.
5) **Label Encoding:** A consistent alphabetical global mapping was applied across all subsets (e.g., akiec=0, ..., vasc=6) to maintain label integrity across the training and inference phases.

III. Pipeline Overview

The end-to-end data mining pipeline is designed to transform raw dermoscopic imagery into a reliable diagnostic tool. The process, illustrated in Fig. 3, proceeds through the following logical phases:

1) **Data Ingestion and Mapping:** Raw HAM10000 images and metadata are loaded, with absolute file paths mapped to the corresponding `image_id` for efficient retrieval.
2) **Partitioning:** A lesion-level split is applied to ensure disjoint sets, preventing data leakage between the 6,987 training, 1,512 validation, and 1,516 test samples.
3) **Imbalance Management:** The training subset undergoes minority class oversampling to reach a 500-sample floor. Simultaneously, class weights are computed from the original distribution to ensure the loss function remains sensitive to rare pathologies.
4) **Training Loop:** A PyTorch-based `DataLoader` (batch size = 32) feeds images into the model. Optimization is handled by the Adam optimizer using a Weighted Cross-Entropy Loss function.
5) **Learning Rate Scheduling:** To ensure convergence and avoid local minima, a Cosine Annealing Learning Rate scheduler is utilized.

6) **Checkpointing and Evaluation:** The model with the lowest validation loss is checkpointed during training. The final performance metrics are then derived from the held-out test set to ensure an unbiased estimate of real-world performance.

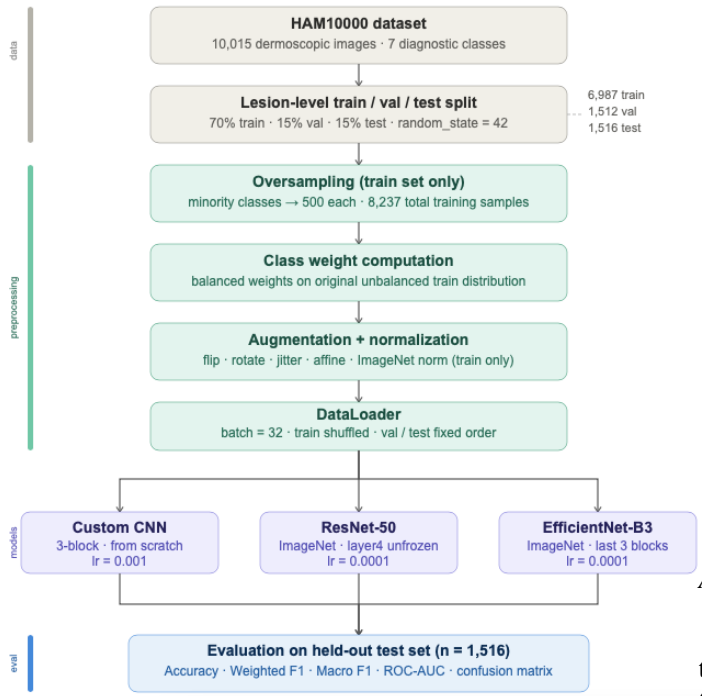

Fig. 3. End-to-end pipeline for HAM10000 skin lesion classification

A. Model Architectures

To evaluate the efficacy of different data mining approaches, three distinct architectures were selected. These models provide a controlled comparison between a from-scratch baseline and transfer learning at varying levels of complexity and parameter depth.

1) **Custom CNN (Baseline):** This architecture serves as a control to evaluate performance without the benefit of prior knowledge from ImageNet. It consists of a three-block convolutional structure:
   - **Block 1:** $\text{Conv2d}(3 \to 32, 3 \times 3, \text{pad} = 1) \to \text{BatchNorm} \to \text{ReLU} \to \text{MaxPool}(2 \times 2)$.
   - **Block 2:** $\text{Conv2d}(32 \to 64, 3 \times 3, \text{pad} = 1) \to \text{BatchNorm} \to \text{ReLU} \to \text{MaxPool}(2 \times 2)$.
   - **Block 3:** $\text{Conv2d}(64 \to 128, 3 \times 3, \text{pad} = 1) \to \text{BatchNorm} \to \text{ReLU} \to \text{AdaptiveAvgPool}(1 \times 1)$.
   - **Classifier:** A fully connected head consisting of a 256-unit hidden layer, Dropout (0.5) for regularization, and a 7-unit output layer.
2) **ResNet-50 (Residual Learning):** Utilizing the `IMAGENET1K_V1` weights, this model employs deep residual learning. To adapt it to dermoscopic images, all parameters were frozen except for the final residual block (`layer4`). This strategy preserves the general edge and texture detectors learned on ImageNet while allowing the high-level feature extractors to specialize in skin lesion morphology.
   - **Trainable Parameters:** 15,491,079 (64.5% of 24.0M total).
   - **Head:** Replaced with a $\text{Dropout}(0.5) \to \text{Linear}(2048, 256) \to \text{ReLU} \to \text{Linear}(256, 7)$ configuration.
3) **EfficientNet-B3 (Compound Scaling):** Selected for its superior accuracy-to-efficiency ratio, this model uses compound scaling to balance depth, width, and resolution. A more aggressive fine-tuning approach was used here by unfreezing the final three feature blocks (`features[-3:]`).
   - **Trainable Parameters:** 8,901,349 (80.3% of 11.1M total).
   - **Head:** Optimized with $\text{Dropout}(0.4) \to \text{Linear}(\text{in_features}, 256) \to \text{ReLU} \to \text{Linear}(256, 7)$.

IV. Experimental Setup and Evaluation

A. Setup

All experiments were conducted using the Google Colaboratory environment equipped with an NVIDIA Tesla T4 GPU (16 GB GDDR6 VRAM) to facilitate accelerated deep learning. The software environment utilized Python 3.10 and the PyTorch 2.x framework. Core libraries included `torchvision` for image transformations, `scikit-learn` for metric computation, and `pandas`/`NumPy` for metadata management.

Table II summarizes the hyperparameter configurations applied across the three models. To ensure a fair comparison, all models were trained for 35 epochs using the Adam optimizer and a Cosine Annealing Learning Rate scheduler. This scheduler oscillates the learning rate following the cosine function, allowing for better convergence toward the end of the training cycle.

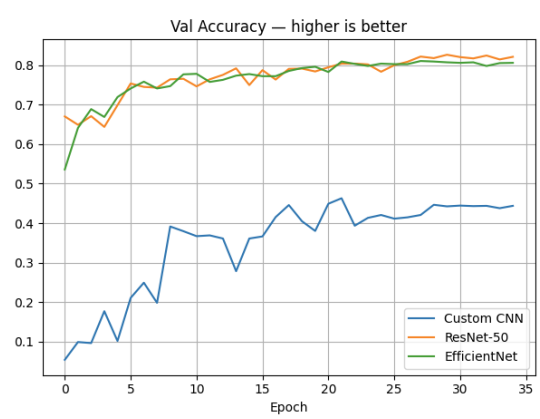

Fig. 4. Validation accuracy across 35 epochs for all three models.

TABLE II
HYPERPARAMETER CONFIGURATION

Hyperparameter	Custom CNN	ResNet-50	EfficientNet-B3
Learning Rate	1×10^{-3}	1×10^{-4}	1×10^{-4}
Optimizer	Adam	Adam	Adam
Scheduler	Cosine Annealing	Cosine Annealing	Cosine Annealing
Epochs	35	35	35
Batch Size	32	32	32
Input Resolution	300×300	300×300	300×300
Trainable Layers	All (Baseline)	`layer4` + Head	`features[-3:]` + Head
Dropout Rate	0.5	0.5	0.4
Loss Function	Weighted CE	Weighted CE	Weighted CE

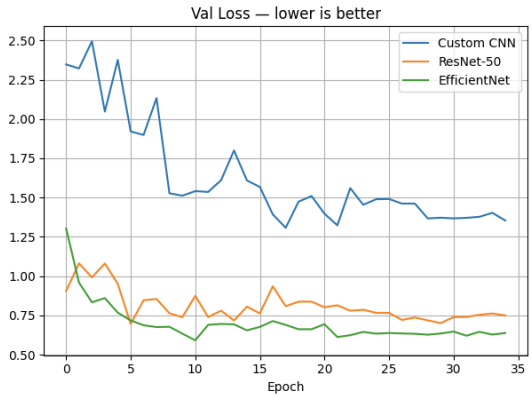

Fig. 5. Validation loss across 35 epochs for all three models.

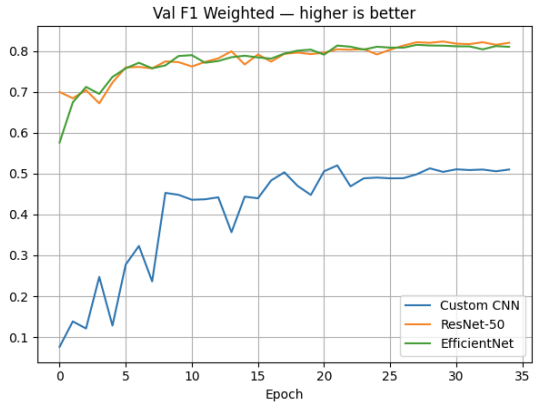

Fig. 6. Validation weighted F1 across 35 epochs for all three models.

Figs. 5, 4, and 6 illustrate the validation loss, accuracy, and weighted F1 across all 35 epochs. ResNet-50 and EfficientNet-B3 converge rapidly after epoch 5, while the Custom CNN exhibits high variance throughout training, confirming its inability to generalize.

B. Evaluation Metrics

Standard accuracy is often an unreliable metric in the presence of extreme class imbalance. Therefore, this study utilizes a multi-metric approach to evaluate model performance from different perspectives:

- **Accuracy:** Calculated as $\frac{TP+TN}{TP+TN+FP+FN}$, providing a general overview of performance.

TABLE III
OVERALL MODEL PERFORMANCE ON TEST SET

Metric	Custom CNN	ResNet-50	EfficientNet-B3	Best Model
Accuracy	0.4261	0.7612	0.7592	ResNet-50
Weighted F1	0.4942	0.7732	0.7797	EfficientNet-B3
Macro F1	0.2215	0.5862	0.6709	EfficientNet-B3
Precision (Wtd)	0.7506	0.7968	0.8202	EfficientNet-B3
Recall (Wtd)	0.4261	0.7612	0.7592	ResNet-50
ROC-AUC (OvR)	0.8458	0.9332	0.9509	EfficientNet-B3

- **Weighted F1-Score:** The harmonic mean of precision and recall, weighted by the number of instances for each label. This accounts for label imbalance by giving more weight to the majority class (`nv`).
- **Macro F1-Score:** The arithmetic mean of all per-class F1-scores. This is our primary metric for clinical relevance, as it treats all seven lesion types with equal importance, regardless of their frequency.
- **ROC-AUC (One-vs-Rest):** Measures the model's ability to distinguish between classes across all possible thresholds. A Macro-AUC is calculated to assess the overall discriminative power.
- **Sensitivity (Recall) for `mel`:** Given that Melanoma is the most life-threatening class, specific focus is placed on its recall to minimize false negatives (missed diagnoses).

V. RESULTS AND ANALYSIS

A. Overall Model Performance Comparison

Table III presents the overall test-set performance ($n = 1,516$) for all three models. EfficientNet-B3 emerged as the superior model for balanced classification, leading in Macro F1 (0.6709), Weighted F1 (0.7797), and ROC-AUC (0.9509). While ResNet-50 achieved a slightly higher raw accuracy (0.7612 vs. 0.7592), it lagged behind EfficientNet-B3 in its ability to recognize minority classes. The Custom CNN baseline underperformed significantly, failing to generalize beyond the majority class.

B. Per-Class Performance Analysis

A granular analysis of per-class metrics reveals how each model handles specific pathologies. Table IV shows that the Custom CNN collapses on most classes — despite high recall for `df` (0.78) and `vasc` (0.88), precision is near zero, confirming the model predicts these classes indiscriminately rather than learning discriminative features.

TABLE IV
PER-CLASS PERFORMANCE — CUSTOM CNN

Class	Precision	Recall	F1-Score	Support
akiec	0.00	0.00	0.00	51
bcc	0.33	0.05	0.09	77
bkl	0.52	0.09	0.16	185
df	0.04	0.78	0.08	23
mel	0.24	0.68	0.35	145
nv	0.96	0.48	0.64	1019
vasc	0.13	0.88	0.23	16

As shown in Fig. 7, the confusion matrix confirms this pattern — the model overwhelmingly predicts nv, with most minority class samples misclassified into nv or mel.

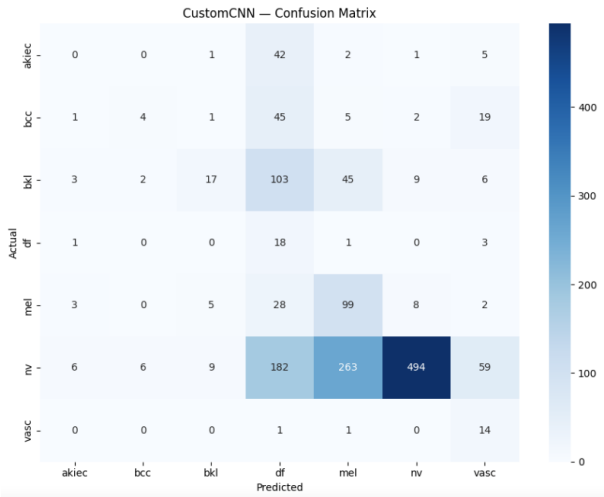

Fig. 7. Confusion matrix — Custom CNN.

C. Per-Class Performance Analysis

A granular analysis of per-class metrics reveals how each model handles specific pathologies. As shown in Table V, ResNet-50 is highly effective at identifying vasc and bcc but exhibits a weakness in detecting mel (recall = 0.51) and df (recall = 0.39).

TABLE V
PER-CLASS PERFORMANCE — RESNET-50

Class	Precision	Recall	F1-Score	Support
akiec	0.37	0.67	0.48	51
bcc	0.61	0.82	0.70	77
bkl	0.56	0.62	0.59	185
df	0.43	0.39	0.41	23
mel	0.48	0.51	0.50	145
nv	0.93	0.83	0.88	1019
vasc	0.39	0.94	0.56	16

The confusion matrix in Fig. 8 shows ResNet-50 correctly identifies the majority of nv samples (844 of 1,019) and vasc (15 of 16), but struggles with df, misclassifying 14 of 23 samples into other classes.

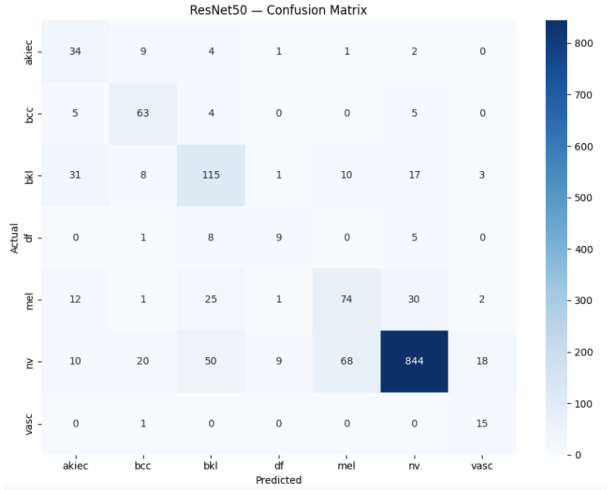

Fig. 8. Confusion matrix — ResNet-50.

In contrast, Table VI highlights the clinical advantage of EfficientNet-B3. It achieved a significant gain in Melanoma (mel) recall (0.66 compared to 0.51), which is the most critical metric for patient safety. Furthermore, its performance on the rarest class, dermatofibroma (df), improved by 22 percentage points, demonstrating that the compound-scaling architecture is more sensitive to the features of minority classes.

TABLE VI
PER-CLASS PERFORMANCE — EFFICIENTNET-B3

Class	Precision	Recall	F1-Score	Support
akiec	0.46	0.65	0.54	51
bcc	0.71	0.71	0.71	77
bkl	0.61	0.68	0.64	185
df	0.67	0.61	0.64	23
mel	0.35	0.66	0.46	145
nv	0.96	0.80	0.87	1019
vasc	0.75	0.94	0.83	16

Fig. 9 demonstrates EfficientNet-B3's improved minority class sensitivity — mel correctly identified 95 of 145 samples compared to ResNet-50's 74, and df recall improved from 9 to 14 correct predictions, directly reflecting the higher Macro F1 score.

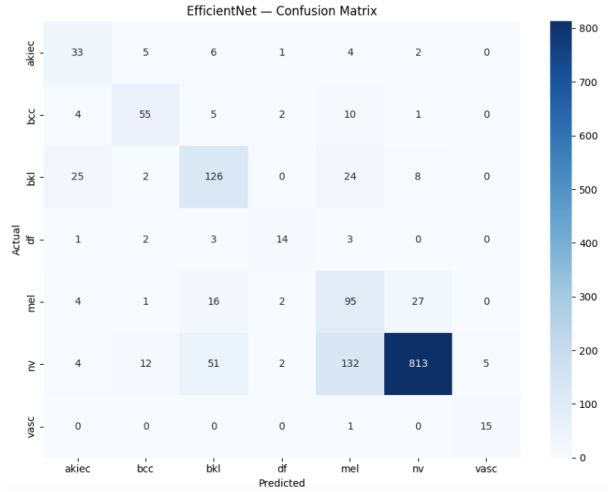

Fig. 9. Confusion matrix — EfficientNet-B3.

D. Analysis of Results

- **Custom CNN Failure:** The baseline model achieved high weighted precision (0.7506) but catastrophic recall (0.4261). This confirms the model essentially defaulted to predicting the majority class (`nv`) for most inputs, failing to learn the underlying features of skin lesions without pre-trained weights.
- **EfficientNet-B3 Advantages:** The 14.5% improvement in Macro F1 over ResNet-50 underscores the efficiency of compound scaling. By scaling depth, width, and resolution simultaneously, the model captures finer dermatological details.
- **The Clinical Trade-off:** EfficientNet-B3 prioritized recall for melanoma (0.66) over precision (0.35). In a clinical setting, this is a favorable trade-off; a false positive results in a biopsy, while a false negative (missing a melanoma) can be fatal.
- **Impact of Imbalance:** Despite oversampling, the gap between Weighted F1 and Macro F1 (0.7797 vs. 0.6709) persists. This quantifies the "imbalance tax"—the difficulty of achieving uniform accuracy across classes when sample sizes range from 16 to 1,019.

VI. Conclusion and Future Work

A. Conclusion

This study conducted a comparative analysis of three deep learning architectures for the classification of skin lesions using the HAM10000 dataset. The results indicate that a from-scratch Custom CNN is insufficient for this complex task (42.61% accuracy), as it lacks the capacity to generalize across severely imbalanced classes without prior feature representations. In contrast, transfer learning models demonstrated significant efficacy. EfficientNet-B3 emerged as the superior architecture for clinical applications, achieving a Macro F1-score of 0.6709, a ROC-AUC of 0.9509, and a critical Melanoma recall of 0.66. While ResNet-50 yielded a marginally higher raw accuracy (76.12%), its lower Macro F1 (0.5862) highlighted a relative inability to recognize rare pathologies. Ultimately, the integration of lesion-level splitting, hybrid imbalance mitigation, and compound-scaled architectures provides a robust framework for automated dermoscopic analysis.

B. Future Work

Future research will focus on enhancing the model's discriminative power and clinical utility:

- **Optimization Strategies:** Implementation of gradual layer unfreezing and learning rate warm-ups to refine the fine-tuning process of EfficientNet.
- **Multi-Modal Fusion:** Integration of patient metadata—such as age, sex, and anatomical location—as auxiliary inputs to provide the model with clinical context.
- **Explainable AI (XAI):** Utilizing Grad-CAM or SHAP-based saliency maps to visualize the diagnostic features prioritized by the model, thereby increasing transparency and physician trust.
- **Generative Augmentation:** Exploring the use of Conditional GANs to generate synthetic samples of rare classes (e.g., dermatofibromas), providing more diverse training data than simple oversampling.

References

[1] P. Tschandl, "The HAM10000 dataset," Kaggle, 2018. [Online]. Available: https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000
[2] N. V. Chawla et al., "SMOTE: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.
[3] D. Bisla et al., "Skin lesion classification: A transfer learning approach using EfficientNets," in *Proc. IEEE CVPR Workshops*, 2019.
[4] E. Nasr-Esfahani et al., "SkinCancerNet: Automated classification of skin lesion using deep transfer learning," *Traitement du Signal*, vol. 40, no. 1, 2023.
[5] M. S. Ali et al., "An efficient deep learning-based skin cancer classifier for an imbalanced dataset," *Diagnostics*, vol. 12, no. 10, 2022.
[6] T. Vuong et al., "Skin lesion classification on imbalanced data using deep learning with soft attention," *Sensors*, vol. 22, no. 19, 2022.
[7] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE CVPR*, 2016, pp. 770–778.
[8] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. ICML*, 2019, pp. 6105–6114.
[9] J. Deng et al., "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE CVPR*, 2009, pp. 248–255.
[10] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. ICLR*, 2015.
[11] I. Loshchilov and F. Hutter, "SGDR: Stochastic gradient descent with warm restarts," in *Proc. ICLR*, 2017.
[12] N. Srivastava et al., "Dropout: A simple way to prevent neural networks from overfitting," *JMLR*, vol. 15, pp. 1929–1958, 2014.
[13] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training," in *Proc. ICML*, 2015, pp. 448–456.